# Augusto B. Corrêa

Junior Research Fellow
Christ Church, Oxford

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Christ Church, St Aldate's, OX1 1DP, Oxford, United Kingdom · Updated: October 9, 2026

## CURRENT POSITION

since 11/2024

**Junior Research Fellow**
at Christ Church, University of Oxford, United Kingdom

## EDUCATION

01/2020 – 10/2024

**Ph.D. in Computer Science**
University of Basel, Switzerland
Thesis: "*Planning with Different Representations*"
*Summa cum laude* (with distinction)
Supervisor: Prof. Dr. Malte Helmert
**Winner of the ICAPS Outstanding Dissertation Award (2025)**

02/2018 – 12/2019

**M.Sc. in Computer Science**
University of Basel, Switzerland
Major in Machine Intelligence

03/2013 – 01/2018

**B.Sc. in Computer Science**
Federal University of Rio Grande do Sul, Brazil
*Winner of the Brazilian Computer Society's Best Bachelor's Thesis Award*

## AWARDS AND HONOURS

2026

**ICAPS 2026 Best Paper Runner-Up Award**
For the paper "*Potential Heuristics as Real-Valued Multilinear Polynomials*"

2026

**Google's Gemini Academic Program Award**
USD 10 000 for research on large language models for planning

2026

**ICAPS 2026 Outstanding Program Committee Member Award**
For service on the ICAPS 2026 program committee

2025

**ICAPS Outstanding Dissertation Award**
For the PhD dissertation "*Planning with Different Representations*"

07/2023

**Winner, IPC 2023 Classical Satisficing Track**
For the planners "*Scorpion Maidu*" and "*Levitron*"
at the 10th International Planning Competition (IPC 2023) at ICAPS 2023

07/2023

**Runner-Up, IPC 2023 Classical Satisficing and Agile Tracks**
For the planner "*Fast Downward Stone Soup 2023*"
at the 10th International Planning Competition (IPC 2023) at ICAPS 2023

07/2023

**ICAPS 2023 Best Student Paper Runner-Up Award**
For the paper "*Grounding Planning Tasks Using Tree Decompositions and Iterated Solving*"

09/2022

**Invited to the Ninth Heidelberg Laureate Forum**
Young Researcher at the 9th Heidelberg Laureate Forum

06/2022

**ICAPS 2022 Best System Demonstration Award**
For the demo "*Machetli: Simplifying Input Files for Debugging*"

10/2019

**Härtefallstipendium – Autumn Semester 2019**
Scholarship for undergraduate students at the University of Basel
Value: 4,000 CHF

09/2019 **Invited to Seventh Heidelberg Laureate Forum**
Young Researcher at the 7th Heidelberg Laureate Forum

05/2019 **Härtefallstipendium – Spring Semester 2019**
Scholarship for undergraduate students at the University of Basel
Value: 1,000 CHF

2018 **Brazilian Computer Society's Award for Best Bachelor's Thesis**
*37º Concurso de Trabalhos de Iniciação Científica (CTIC)*
For the B.Sc. thesis *"Domain-Dependent Heuristics and Tie-Breakers: Topics in Automated Planning"*

11/2017 **ACM ICPC Latin America Finalist**
24th place among 72 teams in the Brazilian regional
Qualified as first-ranked team in Rio Grande do Sul supersite
(19th place among 837 teams in the Brazilian sub-regional)

10/2016 – 08/2017 **Erasmus Mundus Smart$^2$ Programme**
Ten months mobility at Technical University of Munich, Germany
Funded by the European Commission
Value: 10,000 Euros over 10 months

07/2013 – 06/2014 **Young Talents for Science**
*Jovens Talentos para a Ciência*
One year scholarship for freshman students
Funded by the Coordination for the Improvement of Higher Education Personnel (*CAPES*)
Value: 4,800 BRL over one year

## INVITED TALKS

09/2026 **Talk at Sapienza University of Rome**
Seminar at the Department of Computer, Control and Management Engineering (DIAG)
Talk: *Planning in the Age of Large Language Models*

09/2026 **Talk at the Aachen RLeap Symposium 2026**
Invited talk at the Aachen Symposium on Representation Learning to Act and Plan, organized by RWTH Aachen University
Talk: *We Were Wrong About LLMs and Planning*

06/2025 **Talk at Christ Church 500 Years' Alumni Week**
Invited talk at the Alumni Week to celebrate the 500 years of Christ Church college, University of Oxford
Talk: *Not All AI is ChatGPT*

06/2025 **Talk at Linköping University**
Research talk at the Artificial Intelligence and Integrated Computer Systems Division of the Linköping University
Talk: *Planning, LLMs, and the Holy Grail*

10/2024 **Talk at ActSynt Workshop at ECAI 2024**
Invited talk at the Workshop on Highlights of Reasoning about Actions, Planning and Reactive Synthesis, at ECAI 2024
Talk: *Lifted Planning: Recent Advances in Planning Using First-Order Representations*

07/2023 **Talk at King's College London**
Invited talk at King's College London
Talk: *Zero-Knowledge Proofs for Classical Planning*

## RESEARCH VISITS

06/2025 **Research Visit**
Working with Prof. Jendrik Seipp
Linköping University, Sweden

06/2023 – 07/2023 and 01/2024 – 02/2024 **Visiting Student**
Working with Prof. Giuseppe De Giacomo
University of Oxford, United Kingdom

05/2022 **Visiting Student**
Working with Prof. Georg Gottlob's Databases and Artificial Intelligence Group
Technical University of Vienna, Austria

2016 – 2017 **Exchange Student at TU Munich**
Ten month mobility at Technical University of Munich, Germany

03/2017 **Visiting Student**
Department of Mathematics and Computer Science - Artificial Intelligence Group
University of Basel

## RESEARCH EXPERIENCE

01/2020 – 10/2024 **Researcher**
Department of Mathematics and Computer Science - Artificial Intelligence Group
University of Basel
Supervisor: Prof. Dr. Malte Helmert

02/2018 – 12/2019 **Research Assistant**
Department of Mathematics and Computer Science - Artificial Intelligence Group
University of Basel
Supervisor: Prof. Dr. Malte Helmert

07/2014 – 01/2018 **Research Assistant**
Institute of Informatics - Algorithms and Optimization Group
Federal University of Rio Grande do Sul
Supervisors: Prof. Dr. Marcus Ritt and Prof. Dr. André G. Pereira

07/2013 – 06/2014 **Research Assistant**
Institute of Informatics - Data Mining and Data Extraction Group
Federal University of Rio Grande do Sul
Supervisor: Prof. Dr. Renata Galante

## TEACHING EXPERIENCE

01/2025 – 03/2025 **Tutor**
Tutor for the course "Concurrent Programming"
Christ Church, University of Oxford

02/2024 – 06/2024 **Class Tutor**
Tutor for the course "Foundations of Artificial Intelligence"
University of Basel
Lecturer: Prof. Dr. Malte Helmert

09/2022 – 12/2022 **Lecturer**
Lecturer for the seminar "Algorithm Engineering"
University of Basel
(Together with Prof. Dr. Malte Helmert and Dr. Florian Pommerening)

09/2021 – 12/2021 **Lecturer**
Lecturer for the seminar "Theory and Algorithms of Puzzles and Games"
University of Basel
(Together with Prof. Dr. Malte Helmert and Dr. Liat Cohen)

09/2021 – 12/2021 **Lecturer**
Lecturer for the seminar "Scientific Writing"
University of Basel
(Together with Prof. Craig Hamilton)

03/2021 – 06/2021 **Class Tutor**
Tutor for the course "Theory of Computer Science"
University of Basel
Lecturer: Dr. Gabriele Röger

09/2020 – 01/2021 **Lecturer**
Lecturer for the seminar "Turing Award Winners and Their Contributions"
University of Basel
(Together with Prof. Dr. Malte Helmert and Dr. Florian Pommerening)

09/2020 – 01/2021 **Class Tutor**
Tutor for the course "Discrete Mathematics in Computer Science"
University of Basel
Lecturers: Prof. Dr. Malte Helmert and Dr. Gabriele Röger

05/2017 – 07/2017 **Teaching Assistant**
Assistant for the course "Graph Theory and Combinatorics"
Federal University of Rio Grande do Sul
Lecturer: Prof. Dr. André G. Pereira

## PUBLICATIONS

### Journal Articles

1 **Corrêa, A. B.**; Pereira, A. G.; and Seipp, J.
Frontier Large Language Models Are Becoming Competitive with State-of-the-Art Planners.
*Communications of the ACM (CACM)*. Accepted; to appear.

### Peer-Reviewed Papers at Major Conferences

1 Meneguzzi, F.; Buchweitz, A.; **Corrêa, A.B.**; Putrich, V.S.; and Pereira, A.G.
Hierarchical Task Network Planning with LLM-Generated Heuristics.
In *40th Annual Conference on Neural Information Processing Systems (NeurIPS 2026)*. To appear.

2 Pereira, A.G.; **Corrêa, A.B.**; and Seipp, J.
Property-Guided LLM Program Synthesis for Planning.
In *40th Annual Conference on Neural Information Processing Systems (NeurIPS 2026)*. To appear.

3 **Corrêa, A. B.**; Dold, S.; and Helmert, M.
Potential Heuristics as Real-Valued Multilinear Polynomials.
In *36th International Conference on Automated Planning and Scheduling (ICAPS 2026)*.
**ICAPS 2026 Best Paper Runner-Up Award**.

4 **Corrêa, A. B.**; Pereira, A. G.; and Seipp, J.
Classical Planning with LLM-Generated Heuristics: Challenging the State of the Art with Python Code.
In *39th Annual Conference on Neural Information Processing Systems (NeurIPS 2025)*.

5 **Corrêa, A. B.**
Delete-Free Planning with Object Creation is Undecidable.
In *35th International Conference on Automated Planning and Scheduling (ICAPS 2025)*.

6 **Corrêa, A. B.**; Seipp, J.
Alternation-Based Novelty Search.
In *35th International Conference on Automated Planning and Scheduling (ICAPS 2025)*.

7 Speck, D.; Hecher, M.; Gnad, D.; Fichte, J.K.; and **Corrêa, A. B.**
Counting and Reasoning with Plans.
In *39th Annual AAAI Conference on Artificial Intelligence (AAAI 2025)*.

8 **Corrêa, A. B.**; and De Giacomo, G.
Lifted Planning: Recent Advances in Planning Using First-Order Representations.
In *33rd International Joint Conference on Artificial Intelligence (IJCAI 2024)*.

9 **Corrêa, A. B.**; De Giacomo, G.; Helmert, M.; and Rubin, S.
Planning with Object Creation.
In *34th International Conference on Automated Planning and Scheduling (ICAPS 2024)*.

10 **Corrêa, A. B.**; Hecher, M.; Helmert, M.; Longo, D. M.; Pommerening, F. and Woltran, S.
Grounding Planning Tasks Using Tree Decompositions and Iterated Solving.
In *33rd International Conference on Automated Planning and Scheduling (ICAPS 2023)*.
**ICAPS 2023 Best Student Paper Runner-Up Award**.

11 **Corrêa, A. B.**; Büchner, C. and Christen, R.
Zero-Knowledge Proofs for Classical Planning Problems.
In *37th AAAI Conference on Artificial Intelligence (AAAI 2023)*.

12 **Corrêa, A. B.** and Seipp, J.
Best-First Width Search for Lifted Classical Planning.
In *32nd International Conference on Automated Planning and Scheduling (ICAPS 2022)*.

13 Helmert, M.; Sievers, S.; Rovner A. and **Corrêa, A. B.**
On the Complexity of Heuristic Synthesis for Satisficing Classical Planning: Potential Heuristics and Beyond.
In *32nd International Conference on Automated Planning and Scheduling (ICAPS 2022)*.

14 **Corrêa, A.B.**; Pommerening, F.; Helmert, M. and Francès, G.
The FF Heuristic for Lifted Classical Planning.
In *36th AAAI Conference on Artificial Intelligence (AAAI 2022)*.

15 **Corrêa, A.B.**; Francès, G.; Pommerening, F.; and Helmert, M.
Delete-Relaxation Heuristics for Lifted Classical Planning.
In *31st International Conference on Automated Planning and Scheduling (ICAPS 2021)*.

16 **Corrêa, A.B.**; Pommerening, F.; Helmert, M. and Francès, G.
Lifted Successor Generation using Query Optimization Techniques.
In *30th International Conference on Automated Planning and Scheduling (ICAPS 2020)*.

17 Francès, G.; **Corrêa, A.B.**; Geissmann, C. and Pommerening, F.
Generalized Potential Heuristics for Classical Planning.
In *28th International Joint Conference on Artificial Intelligence (IJCAI 2019)*.

18 **Corrêa, A.B.** and Pommerening, F.
An Empirical Study of Perfect Potential Heuristics.
In *29th International Conference on Automated Planning and Scheduling (ICAPS 2019)*.

19 **Corrêa, A.B.**; Pereira, A. G. and Ritt, M.
Analyzing Tie-Breaking Strategies for the A* Algorithm.
In *27th International Joint Conference on Artificial Intelligence (IJCAI 2018)*.

### Planners and System Demos

1 **Corrêa, A. B.**; Francès, G.; Hecher, M.; Longo, D. M. and Seipp, J.
Scorpion Maidu: Width Search in the Scorpion Planning System.
In *10th International Planning Competition (IPC 2023) Classical Tracks*.
**Winner, IPC 2023 Classical Satisficing Track**.

2 **Corrêa, A. B.**; Francès, G.; Hecher, M.; Longo, D. M. and Seipp, J.
Levitron: Combining Ground and Lifted Planning.
In *10th International Planning Competition (IPC 2023) Classical Tracks*.
**Winner, IPC 2023 Classical Satisficing Track**.

3 **Corrêa, A. B.**; Francès, G.; Hecher, M.; Longo, D. M. and Seipp, J.
The Powerlifted Planning System in the IPC 2023.
In *10th International Planning Competition (IPC 2023) Classical Tracks*.

4 Doebber, D; Pereira, A. G., and **Corrêa, A. B.**.
OpCount4Sat: Operator Counting Heuristics for Satisficing Planning.
In *10th Planning Competition (IPC 2023) Classical Tracks*.

5 Büchner, C.; Christen, R.; **Corrêa, A.B.**; Eriksson, S.; Ferber, P.; Seipp, J. and Sievers, S.
Fast Downward Stone Soup 2023.
In *10th International Planning Competition (IPC 2023) Classical Tracks*.
**Runner-Up, IPC 2023 Classical Satisficing Track**.
**Runner-Up, IPC 2023 Classical Agile Track**.

6 Galery Käser, L.; Büchner, C.; **Corrêa, A.B.**; Pommerening, F.; and Röger, G.
Machetli: Simplifying Input Files for Debugging.
In *System Demonstrations at the 32nd International Conference on Automated Planning and Scheduling (ICAPS 2022)*.
**ICAPS 2022 Best System Demonstration Award**.

### Other Publications

1. **Corrêa, A.B.**; Gelberg Y.; Melo, L.C.; Shumailov, I.; Pereira, A.G.; and Gal, Y.
Iterative Deployment Improves Planning Skills in LLMs.
arXiv:2512.24940.
2. Haslum, P.; and **Corrêa, A.B.**.
The Universal PDDL Domain.
arXiv:2411.08040, 2024.
3. **Corrêa, A.B.**; and Seipp, J.
Consolidating LAMA with Best-First Width Search.
In *ICAPS 2024 Workshop on Heuristics and Search for Domain-independent Planning (HSDIP 2024)*.
4. Abdulaziz, M.; Pommerening, F.; and **Corrêa, A.B.**
Mechanically Proving Guarantees of Generalized Heuristics: First Results and Ongoing Work.
In *ICAPS 2022 Workshop on Heuristics and Search for Domain-independent Planning (HSDIP 2022)*.
5. De Graaff, R; **Corrêa, A.B.** and Pommerening, F.
Concept Languages as Expert Input for Generalized Planning: Preliminary Results.
In *ICAPS 2021 Workshop on Knowledge Engineering for Planning and Scheduling (KEPS 2021)*.
6. **Corrêa, A.B.**; Pommerening, F. and Francès, G.
Relaxed Decision Diagrams for Delete-Free Planning.
In *CP-2018 Workshop on Constraints & AI Planning*.
7. **Corrêa, A.B.**; Pereira, A. G. and Ritt, M.
Domain-Dependent Heuristics and Tie-Breakers: Topics in Automated Planning.
In *Revista Eletrônica de Iniciação Científica (REIC)*.
8. **Corrêa, A.B.**; Pereira, A. G. and Ritt, M.
Improved Airport Ground Traffic Control with Domain-Dependent Heuristics.
In *5th Brazilian Conference on Intelligent Systems (BRACIS 2016)*.

## ACADEMIC SERVICE

### Journals

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|---|---|
| AIJ | Reviewer (2023, 2024, 2026) |
| JAIR | Reviewer (2022, 2024, 2025) |

### Conferences, Workshops, and Competitions

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| AAAI | PC Member (2021–2024), PC Member Demo Track (2025, 2026) |
| ECAI | PC Member (2022, 2024) |
| IJCAI | PC Member (2022, 2023, 2025), Reviewer (2019) |
| ICAPS | PC Member (2021, 2024–2026), Reviewer (2018–2020, 2022, 2023) |
| KR | PC Member (2024, 2025) |
| ICLR | Reviewer (2026, 2027) |
| LM4Plan Workshop | Organizer (ICAPS 2025, 2026; ICML 2026) |
| NeurIPS | PC Member (2025) |
| SoCS | PC Member (2023, 2025), Reviewer (2021, 2022) |
| IPC | Assistant Organizer (2018) |

### Expert Panels

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|---|---|
| LEAP | Transformative AI expert on the Longitudinal Expert AI Panel of the Forecasting Research Institute (since Wave 12, 2026) |

## SUPERVISION

### As Supervisor: Completed

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| 11/2024 | **Yash Trivedi (B.Sc.)**<br>*Recasting Stackelberg Planning Problems as FOND Planning* |

04/2023 **Travis Rivera Petit (M.Sc.)**
*Computational Complexity of Classical Planning Domains Based on Grids*

04/2023 **Nícolas Duranti (B.Sc.)**
*A Non-Admissible Heuristic Function Based on Synchronized Abstract Plans*
Co-supervision with Prof. André Grahl Pereira.

07/2022 **Sebastian Schlachter (B.Sc.)**
*Encoding Diverse Sudoku Variants as SAT Problems*

06/2022 **Ken Rotaris (B.Sc.)**
*Single-Player Chess as a Planning Problem*

06/2021 **Simon Dold (M.Sc.)**
*Correlation Complexity and Different Notions of Width*
Co-supervision with Thomas Keller.

10/2020 **Lucas Galery Käser (B.Sc.)**
*Finding Small Counterexamples of Expected Planner Behavior with Hill-Climbing*
Co-supervision with Gabriele Röger.

06/2020 **Rik De Graaff (B.Sc.)**
*Concept Languages as Expert Input for Generalized Planning*
Co-supervision with Florian Pommerening.

### As External Examiner

04/2025 **Frederico Messa Schwartzhaupt**
*FOND Planning via Explicit Search.*

## CONTRIBUTIONS TO OPEN SOURCE PROJECTS

- Powerlifted: Planning System implemented in C++ and Python (creator)
- pddl-skills: Skills and Python tools for PDDL planning with LLMs (creator)
- llm-generated-heuristics: Code for the NeurIPS 2025 paper on LLM-generated heuristics (creator)
- asp-grounding-planning: Grounding via logic programming; code for the ICAPS 2023 paper (creator)
- Fast Downward: Planning System implemented in C++ and Python
- Prost: Probabilistic Planning System implemented in C++
- Downward Lab: Experiment Framework implemented in Python
- PlanPilot: Tool to interactively navigate the plan space of planning tasks
- htg-domains: Collection of hard-to-ground planning domains (maintainer)
- object-creation-benchmarks: PDDL domains from the ICAPS 2024 paper on object creation (creator)

## LANGUAGES

Portuguese (native), English (fluent), German (advanced), Spanish (advanced).